

Econ 210B

Discussion Session N°3

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Plan for Today

Some Definitions

- ▶ **System of Ordinary Differential Equations (ODE):** Set of equations involving multiple interdependent variables and their rate of change to a single independent variable (typically t). Example:

$$\dot{x}_1(t) = f_1(x_1(t), x_2(t), \dots, x_n(t), t)$$

$$\dot{x}_2(t) = f_2(x_1(t), x_2(t), \dots, x_n(t), t)$$

$$\vdots$$

$$\dot{x}_n(t) = f_n(x_1(t), x_2(t), \dots, x_n(t), t)$$

Where $\dot{x}_i(t) \equiv \frac{dx_i(t)}{dt}$ for each $i = 1, 2, \dots, n$.

- ▶ **Phase Diagram:** Used to visualize the behavior of an ODE system (typically with 2 variables). Includes:
 - ▶ **Loci:** Points where one variable is at rest, i.e. $\dot{x}_i(t) = 0$.
 - ▶ **Regions:** Each locus divides the graph in 2 regions with respect to the sign of $\dot{x}_i(t)$.
 - ▶ **Winds:** To illustrate change of the variable in x-axis (y-axis) we use $\rightarrow$ $\leftarrow$ ($\downarrow$ $\uparrow$)

Building a Phase Diagram

Think about a general 2 variable system (y and x) defined by the ODE:

$$\dot{x}(t) = ax + by + h$$

$$\dot{y}(t) = cx + dy + k$$

1. **Loci:** graph the points (x, y) at which (1) $\dot{x}(t) = 0$ and (2) $\dot{y}(t) = 0$.

$$\dot{x}(t) = 0 \iff ax + by + h = 0 \iff y = -\frac{a}{b}x - \frac{h}{b}$$

$$\dot{y}(t) = 0 \iff cx + dy + k = 0 \iff y = -\frac{c}{d}x - \frac{k}{d}$$

2. **Winds:** Characterize the behavior of the system in each area divided by (1) and (2) separately.
3. **System Dynamics:** Put (1) + (2) + Winds together and you have characterized the dynamics of the system.
4. **Stability:** System can be globally stable, unstable or saddle-path stable.

Building a Phase Diagram: (1) $\dot{x}(t) = 0$

1. **Loci:** graph the points (x, y) at which (1) $\dot{x}(t) = 0$.

$$\dot{x}(t) = 0 \iff ax + by + h = 0 \iff y = -\frac{a}{b}x - h$$

2. **Winds:** Characterize the behavior of the system in each area divided by (1).

Assume $a, c, d < 0$, $b > 0$.

Building a Phase Diagram: (1) $\dot{x}(t) = 0$

1. **Loci:** graph the points (x, y) at which (1) $\dot{x}(t) = 0$.

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2. **Winds:** Characterize the behavior of the system in each area divided by (1).

→ Hint: Take (x, y) as given, then move y and see what happens to $\dot{x}$.

Assume $a, c, d < 0$, $b > 0$.

Building a Phase Diagram: (2) $\dot{y}(t) = 0$

1. **Loci:** graph the points (x, y) at which (2) $\dot{y}(t) = 0$.

$$\dot{y}(t) = 0 \iff cx + dy + k = 0 \iff y = -\frac{c}{d}x - k$$

2. **Winds:** Characterize the behavior of the system in each area divided by (2).

Assume $a, c, d < 0$, $b > 0$.

Building a Phase Diagram: (2) $\dot{y}(t) = 0$

1. **Loci:** graph the points (x, y) at which (2) $\dot{y}(t) = 0$.

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2. **Winds:** Characterize the behavior of the system in each area divided by (2).

→ Hint: Take (x, y) as given, then move x and see what happens to $\dot{y}$.

Assume $a, c, d < 0$, $b > 0$.

Building a Phase Diagram: System Dynamics

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3. **System Dynamics:** Put (1) + (2) + Winds together and you have characterized the dynamics of the system.

Example: Neoclassical Growth Model

$$\begin{aligned} \max \quad & \int_0^{\infty} e^{-\rho t} \frac{c(t)^{1-\eta} - 1}{1-\eta} dt \\ \text{s.t.} \quad & \dot{k}(t) = Af(k(t)) - \delta k(t) - c(t), \\ & k(0) = k_0 > 0, \\ & k(t) \geq 0 \end{aligned}$$

$$\tilde{\mathcal{H}}(t) = \frac{c(t)^{1-\eta} - 1}{1-\eta} + \mu(t)[Af(k(t)) - \delta k(t) - c(t)]$$

Conditions for an optimum

$$(c) \quad c(t)^{-\eta} - \mu(t) = 0$$

$$(k) \quad \dot{\mu}(t) - \rho\mu(t) = -\mu(t)[Af'(k(t)) - \delta]$$

$$(\mu) \quad \dot{k}(t) = Af(k(t)) - \delta k(t) - c(t)$$

$$TC \quad \lim_{T \rightarrow \infty} e^{-\rho T} \mu(T)k(T) = 0$$

Example: Neoclassical Growth Model

From

$$(\dot{c}) \quad c(t)^{-\eta} - \mu(t) = 0$$

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We can arrive to an ODE system in $c(t)$ and $k(t)$:

$$(\dot{k}) \quad \dot{k}(t) = Af(k(t)) - \delta k(t) - c(t)$$

$$(\dot{c}) \quad \frac{\dot{c}(t)}{c(t)} = \frac{1}{\eta} (Af'(k(t)) - \rho - \delta)$$

With these 2 equations we will follow the same procedure to characterize the Phase Diagram.

Phase Diagram in Neoclassical Model

$$(\dot{k}) \quad \dot{k}(t) = Af(k(t)) - \delta k(t) - c(t)$$

$$(\dot{c}) \quad \frac{\dot{c}(t)}{c(t)} = \frac{1}{\eta} \left(Af'(k(t)) - \rho - \delta \right)$$

1. **Loci:** graph the points (k, c) at which (1) $\dot{k}(t) = 0$ and (2) $\dot{c}(t) = 0$.

$$\dot{k}(t) = 0 \iff c(t) = Af(k(t)) - \delta k(t)$$

$$\dot{c}(t) = 0 \iff Af'(k(t)) = \rho + \delta$$

2. **Winds:** Characterize the behavior of the system in each area divided by (1) and (2) separately.
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Neoclassical Model Phase Diagram: $\dot{k}(t) = 0$

$$(\dot{k}) \quad \dot{k}(t) = Af(k(t)) - \delta k(t) - c(t)$$

1. **Loci:** graph the points (k, c) at which (1) $\dot{k}(t) = 0$.

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2. **Winds:** Characterize the behavior of the system in each area divided by (1).

→ Hint: Take (k, c) as given, then move c and see what happens to $\dot{k}$.

Building a Phase Diagram: $\dot{c}(t) = 0$

$$(\dot{c}) \quad \frac{\dot{c}(t)}{c(t)} = \frac{1}{\eta} \left(Af'(k(t)) - \rho - \delta \right)$$

1. **Loci:** graph the points (k, c) at which (2) $\dot{c}(t) = 0$.

$$\dot{c}(t) = 0 \iff Af'(k(t)) = \rho + \delta$$

2. **Winds:** Characterize the behavior of the system in each area divided by (2).

Building a Phase Diagram: $\dot{c}(t) = 0$

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→ Hint: Take (k, c) as given, then move k and see what happens to $\dot{c}$.

Building a Phase Diagram: System Dynamics

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3. **System Dynamics:** Put (1) + (2) + Winds together and you have characterized the dynamics of the system.

Plan for Today

PS1. D. Capital Accumulation and Natural Resources

Consider an economy with an aggregate output technology that depends on human-made physical capital, k , and energy from a natural resource, z . The technology is:

$$y(t) = k(t)^\eta z(t)^{1-\eta},$$

where $\eta \in (0, 1)$.

At each instant, capital changes according to the law of motion:

$$\dot{k}(t) = y(t) - \delta k(t) - c(t),$$

where $c(t)$ represents the flow of output consumed at t , $\delta > 0$, and $k(0) = k_0 > 0$.

The social planner aims to maximize the objective function:

$$U = \int_0^\infty c(t) e^{-\rho t} dt,$$

where $\rho > 0$, subject to:

$$c(t) \geq \underline{c}, \quad \forall t \geq 0,$$

and $\underline{c} \geq 0$ is a minimal subsistence consumption level.

Assume the natural resource is renewable, with a maximal flow $\bar{z} > 0$:

$$z(t) \leq \bar{z}, \quad \forall t \geq 0.$$

PS1. D. Hamiltonian and Optimality Conditions

(a) Hamiltonian and Lagrangian:

$$\tilde{\mathcal{H}} = c(t) + \mu(t) [k(t)^\eta z(t)^{1-\eta} - \delta k(t) - c(t)]$$

with additional constraints incorporated into the Lagrangian:

$$\tilde{\mathcal{L}} = c(t) + \mu(t) [k(t)^\eta z(t)^{1-\eta} - \delta k(t) - c(t)] + \theta_1(t)(c(t) - \underline{c}) + \theta_2(t)(\bar{z} - z(t)).$$

- *State variable:* $k(t)$
- *Control variables:* $c(t), z(t)$

(b) Optimality Conditions:

$$(\underline{c}) \quad 1 - \mu(t) + \theta_1(t) = 0 \quad \implies \quad \mu(t) = \theta_1(t) + 1$$

$$(\underline{z}) \quad \mu(t)(1 - \eta)k(t)^\eta z(t)^{-\eta} - \theta_2(t) = 0$$

$$(\dot{k}) \quad \dot{\mu}(t) - \rho\mu(t) = -\tilde{\mathcal{L}}_k = \mu(t) [\delta - \eta k(t)^{\eta-1} z(t)^{1-\eta}]$$

$$(\dot{\mu}) \quad \dot{k}(t) = k(t)^\eta z(t)^{1-\eta} - \delta k(t) - c(t)$$

$$(\theta_1) \quad \theta_1(t)(c(t) - \underline{c}) = 0$$

$$(\theta_2) \quad \theta_2(t)(\bar{z} - z(t)) = 0$$

PS1. D. Characterize steady state

- ▶ Question (c) asks you to characterize a steady state for z , k , c , that is, points at which $\dot{z} = 0$, $\dot{k} = 0$, $\dot{c} = 0$

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$$\dot{k}(t) = 0 \iff c(t) = k(t)^\eta \bar{z}^{1-\eta} - \delta k(t)$$

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$$\text{while } \mu(t) > 1 \implies c(t) = \underline{c}$$

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- ▶ Note also that $\dot{\mu}(t) = 0 \iff \frac{\dot{\mu}(t)}{\mu(t)} = \rho + \delta - \eta k(t)^{\eta-1} \bar{z}^{1-\eta}$

- ▶ So if $k(0) < \bar{k}$, $\dot{\mu} < 0$: agent chooses subsistence consumption level, accumulates capital and watches $\mu \rightarrow 1$

PS1. D. Phase Diagram on the Space (k, μ)

The behavior of the co-state variable $\mu(t)$ determines the following conditions:

$$\mu(t) \left\{ \begin{array}{ll} > 1 \quad \implies \quad c(t) = \underline{c}, \quad \dot{k} = k(t)^\eta \bar{z}^{1-\eta} - \delta k(t) - \underline{c} \\ & \dot{k} = 0 \quad \iff \quad k(t)^\eta \bar{z}^{1-\eta} = \delta k(t) + \underline{c} \\ = 1 \quad \implies \quad c(t) = \bar{c} \\ & \dot{k} = 0 \quad \iff \quad k^{ss} = \bar{z} \left(\frac{\delta + \rho}{\eta} \right)^{1/(\eta-1)} \end{array} \right.$$

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The behavior of the co-state variable $\mu(t)$ determines the following conditions:

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Great! Now follow the steps:

1. **Loci:** graph the points (k, μ) at which (1) $\dot{k}(t) = 0$ and (2) $\dot{\mu}(t) = 0$.
2. **Winds:** Characterize the behavior of the system in each area divided by (1) and (2) separately.
3. **System Dynamics:** Put (1) + (2) + Winds together and you have characterized the dynamics of the system.
4. **Stability:** System can be globally stable, unstable or saddle-path stable.

PS1. D. (1) $\dot{k} = 0$

$$\mu(t) \left\{ \begin{array}{ll} > 1 \implies c(t) = \underline{c}, \quad \dot{k} = k(t)^\eta \bar{z}^{1-\eta} - \delta k(t) - \underline{c} \\ \dot{k} = 0 \iff k(t)^\eta \bar{z}^{1-\eta} = \delta k(t) + \underline{c} \\ \\ = 1 \implies c(t) = \bar{c} \\ \dot{k} = 0 \iff k^{ss} = \bar{z} \left(\frac{\delta + \rho}{\eta} \right)^{1/(\eta-1)} \end{array} \right.$$

1. **Loci:** graph the points (k, μ) at which (1) $\dot{k}(t) = 0$.
2. **Winds:** Characterize the behavior of the system in each area divided by (2).

PS1. D. (1) $\dot{k} = 0$

1. **Loci:** graph the points (k, μ) at which (1) $\dot{k}(t) = 0$.
2. **Winds:** Characterize the behavior of the system in each area divided by (2).

PS1. D. (2) $\dot{\mu} = 0$

1. **Loci:** graph the points (k, μ) at which (2) $\dot{\mu}(t) = 0$.

$$\dot{\mu}(t) = 0 \iff \mu(t) = 1$$

2. **Winds:** Characterize the behavior of the system in each area divided by (2).

PS1. D. (1) + (2) + Winds = Phase Diagram